VERSION CONTROLLING & WORKFLOWS WITH GIT - II

PROGRAMMING APPLICATIONS AND FRAMEWORKS (IT3030)

LEARNING OUTCOMES

- After completing this lecture, you will be able to,
  - Apply the basic concepts of Git for version controlling
  - Apply an efficient branching strategy for a team
  - Describe what a git workflow is and why a workflow is important.
  - Apply a simple git workflow to your own projects.

CONTENTS

- Recap from the previous week
- Git branching
  - Workflows (Branching strategies)
    - GitFlow
    - GitHub Flow
    - Pull Requests (PRs)
    - Best practices for branching
- Git general best practices
- Additional topics
- Summary

FROM THE PREVIOUS WEEK

- Version Controlling: Scenario Discussion
- A bit of history on Version Controlling Systems (VCS)
- Git Basic Concepts
  - Repositories – Local and Remote
  - Each user has their own local repository
    - This local repository can be obtained by,
      - Copying from another user (P2P)
      - **Cloning** it from a remote repository (on GitHub, BitBucket etc.)
      - Initializing their own repository.
  - Commits
  - Basic git operations (git add, commit, push, pull)

GIT BRANCHING: THE BASICS

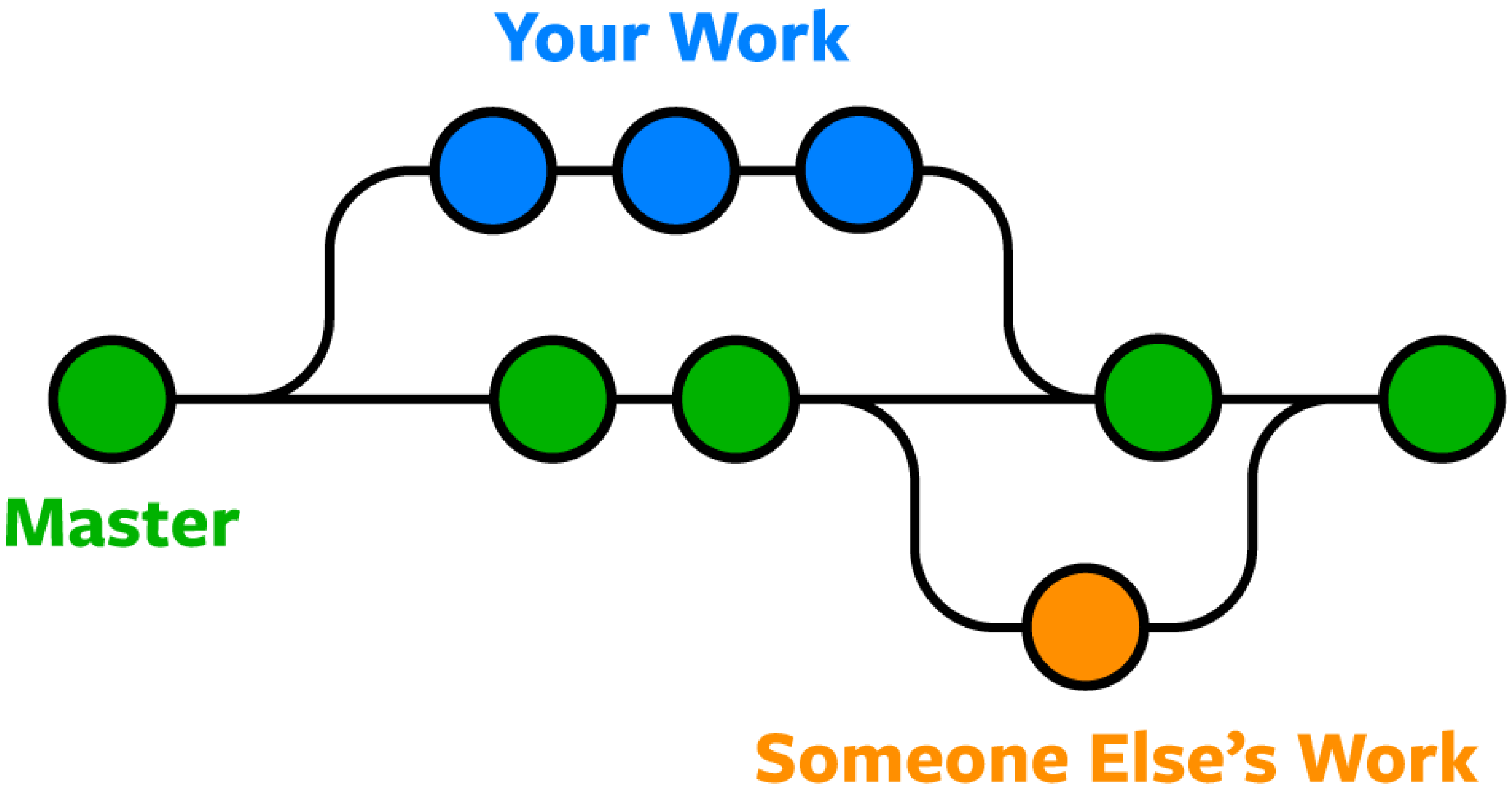
- Think of a tree. What is a branch in a tree?
  - An offshoot of the trunk of a tree.
- What is a branch in Git?
  - The default branch in Git is the ***master/ main*** branch (It is like the trunk of a tree).
  - Very simply put, a branch is a detour you make from the main path of development.

Source: https://www.rainforest-alliance.org/wp-content/uploads/2021/06/kapok-tree-profile-1.jpg.optimal.jpg

GIT BRANCHING: THE BASICS

- Why is branching needed?
  - Branching lets developers work independently inside a single code repository.
    - E.g., In product X, Amith works on feature A, while Binesh works on feature B.
    - If both have their code in just their local repositories, then no problem.
    - Now, it is a good practice to push your changes to the remote repository regularly.
    - What sort of problems can be expected when they do that?
      - **Merge Conflicts!**
  - Any solution to the above problem?
    - Yes, let each of them work on their own branch.

GIT BRANCHING: THE BASICS



Example on git branching. Source: https://www.nobledesktop.com/learn/git/git-branches

GIT BRANCHING: THE BASICS

- Why is branching needed? (continued)
  - Allows a developer to switch among different tasks easily.
    - Suppose while Amith is working on feature A, he gets a request to attend to a major issue in the product X.
    - If Amith is working on his local master branch, he will now have to,
      - Undo all the changes he did.
      - Fix the issue.
      - Restart work on feature A.

GIT BRANCHING: THE BASICS

- If Amith had been working on a separate branch even if he was on his local repository, he could have,
  - Stopped the work on the feature and switch back to master/ main.
  - Created new branch for the **"hotfix"** and work on it.
  - Pushed the hotfix branch to the remote repo (and merge to remote master).
  - Switched back to his feature branch and continue where he left off.

GIT BRANCHING: THE BASICS

- Referring to the previous scenario,
  - Using git branches during development time is clearly more efficient rather than not using any.
  - Using branches is a must for collaborations with multiple developers.
  - Whatever you do on a branch is isolated from the rest of the code – great when you want to experiment!
  - It is not a bad idea to use branching for individual projects too.
- Branching on Git is a very lightweight operation. Therefore, use it whenever possible.

HOW DO WE CREATE A BRANCH ON GIT?

- Very simple.
  - *git branch <branch_name>* - just creates the branch based on your current branch.
  - *git checkout <branch_name>* - checks out the branch.

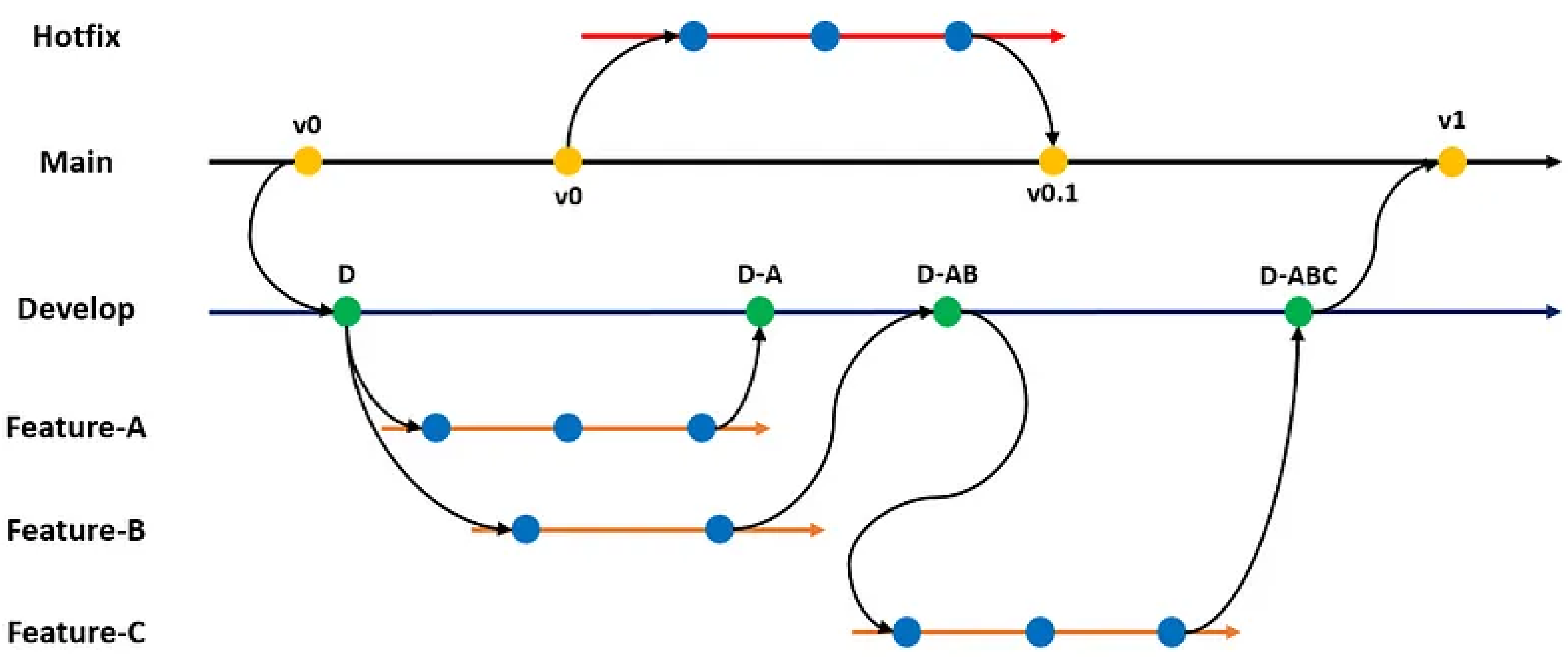
OR

  - *git checkout –b <branch_name>* - creates the branch, and switches to the new branch (checks it out) from your current branch immediately.

WORKFLOWS (BRANCHING STRATEGIES)

- Git is very flexible when it comes to how-to branch. There is no one standard method, but some well accepted how-to branch 'recipes' are there.
- They are called workflows or branching strategies.
  - A Git workflow is a recipe or recommendation for how to use Git to accomplish work in a consistent and productive manner.
- Some popular workflows are,
  - **GitFlow**
  - **GitHub Flow**
  - GitLab Flow
  - Trunk Based Development (goes with CI/ CD)

GITFLOW WORKFLOW



Source: https://betterprogramming.pub/git-branching-strategy-for-better-team-collaboration-aacb5f235d05

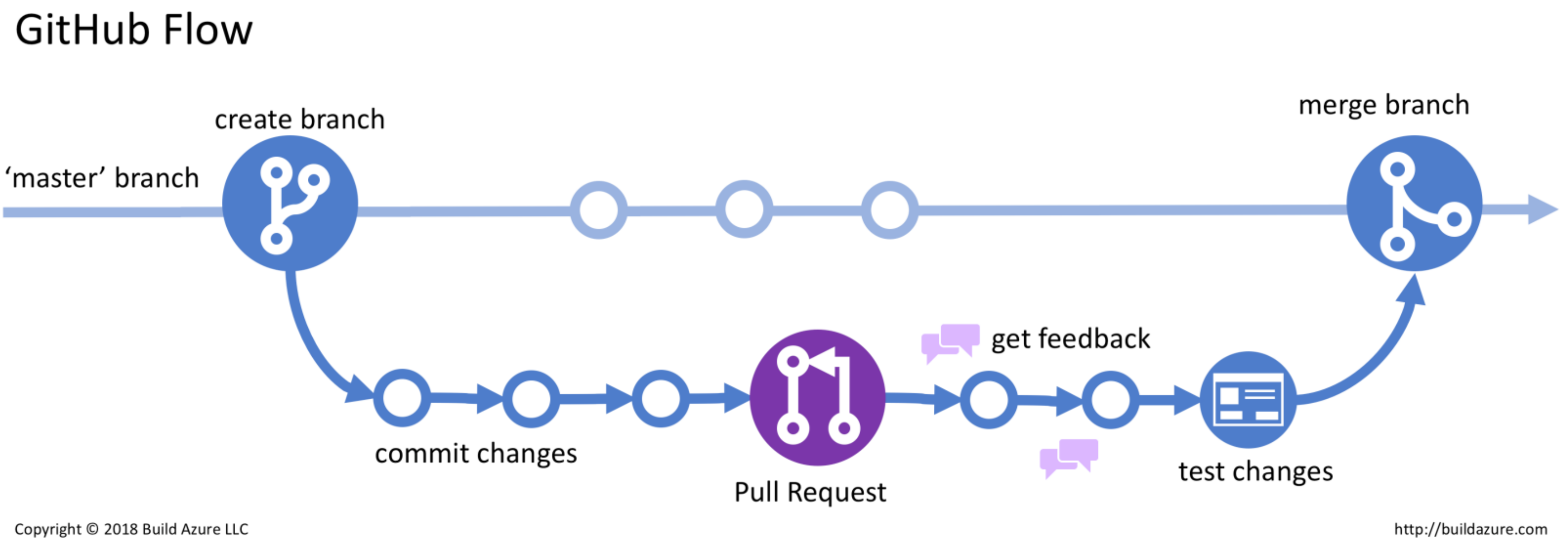
GITFLOW WORKFLOW

- GitFlow is very comprehensive.
  - [YouTube] GitFlow Workflow
  - Can become unnecessarily complex for small projects.
- It is more suited to Enterprise grade software development efforts.
  - Requires enforcement of the practices for the proper functioning of the workflow.
  - For most use cases, it is overkill.

GITHUB WORKFLOW

- A simpler alternative is preferable for most use cases.
  - One such workflow is **GitHub Flow** which can be used by anybody.
  - [YouTube] GitHub Flow

THE GITHUB FLOW



Source: https://build5nines.com/introduction-to-git-version-control-workflow/

GITHUB FLOW

- This is a GitHub based simple workflow.
  1. Create a new feature branch from the main (master) branch.
  2. Make the necessary changes to the feature branch – continue until you think your feature is ready.
  3. Once you think the feature is done, ask for feedback from the other developers – create a Pull Request (PR) for this.
  4. Address their review comments received on the Pull Request (PR).
  5. Merge your Pull Request (PR) to the main branch.
  6. Delete your feature branch.

Refer the official documentation here.

PULL REQUESTS (PR)

- A Pull Request (PR) is used to propose a new change to a repository (usually to the main branch).
- Usually, the proposed changes are in a feature branch, which ensures that the main branch will only contain code which is tested and working.
- The other collaborators will have to verify that the proposed change (PR) is OK first.
- If the PR is OK, they will accept it. Or else, they will suggest changes until it is acceptable.
- Once the PR is accepted, the changes in the feature branch are merged automatically to the main branch, bringing the proposed change to the main branch.
- Find the official documentation on Creating PRs here.

GIT BRANCHING: BEST PRACTICES

1. Create a branch for each feature. Do not work on multiple features in a single branch.
2. Ideally, only one person should work on one feature branch.
3. Feature branches should be short-lived. They **should not exist** once the feature is complete.
4. Delete the local and remote feature branch after merging to the master branch (see point 3).

GIT BRANCHING: BEST PRACTICES

5. Merge early and often to avoid merge conflicts.
6. Keep the master branch clean. Only *production ready* code should be in the master branch. It is better to **avoid** working directly on the master branch.
7. Use a good naming convention to name your feature branches. Everyone should stick to the same naming convention.
   - feature/<username>/task-name-in-lowercase-spaces-replaced-with-dashes
     - feature/vishan.j/awesome-feature-x-to-our-app

GIT GENERAL BEST PRACTICES

1. Make small, incremental changes.
   - Don't make huge changes in one go. If you have a big feature to add, break it down to smaller features and add one at a time.
2. Keep commits atomic.
   - It is better if each commit is focused on one part of the feature. This will minimize the damage if you must revert a commit.
3. Commit often.
   - Commit your changes often. Even if the work is not done.
4. Pull the changes in origin/main branch before you create a new branch from your local/main.
   - This ensures that the new feature branch is created with all the latest changes in master. This reduces the possibility of merge conflicts later.

GIT GENERAL BEST PRACTICES

5. Push local feature branch commits to Remote often.
   - Even if it is an unfinished feature, push it. That's good for safekeeping.
6. Use branches.
   - See the section on branching for a comprehensive set of reasons.
7. Write good commit messages.
   - Make the lives of other Software Engineers, DevOps engineers, QA engineers easier.
8. Select one branching strategy (Git Workflow) and stick with it.
   - Select one that suits you and use it well.

SUMMARY

- Git branching
  - The basics
  - How do we create a branch on git?
  - Workflows (Branching strategies)
    - GitFlow
    - The GitHub Flow
    - Pull Requests (PRs)
  - Best practices for branching
- Git general best practices

ADDITIONAL TOPICS

Read on these topics to be up to date on some current trends.

- Continuous Integration and Continuous Deployment (CI/ CD).
- Compare Trunk based development workflow with GitFlow workflow.
  - Understand when to use which.
- Read on Pull Requests.
- Read on Git Rebase and see why it’s a better alternative to Git Merge.

REFERENCES

1. Pro Git - Scott Chacon and Ben Straub [Free eBook]
2. Dangit, Git!?! [Git quick reference]
3. Branch in Git
4. What Are the Best Git Branching Strategies
5. 10 Git Branching Strategy Best Practices
6. Trunk-based Development vs. Git Flow
7. How to Use Git and Git Workflows – a Practical Guide

THANK YOU

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